TransAtlantic Airlines: Judy's booking-policy problem

Read the case first: `transatlantic-case.pdf`. Build this notebook one step at a time: **ask Claude Code for one step, read what it did, check it makes sense, then ask for the next step.**

Step 1: Framing the decision

Decision: how many reservations to accept per flight on the Heathrow–JFK route, set separately for economy (current limit 420, capacity 400) and business (current limit 120, capacity 100).

Main objective

Minimise the expected cost per flight, defined as lost revenue on empty seats plus the cost of bumping:

- **Empty seat:** costs the ticket price (£450 economy, £1,450 business). No-shows are refunded and the marginal cost of carrying a passenger is negligible, so an empty seat is pure lost revenue.
- **Bumped passenger:** costs £150 (economy) or £300 (business), covering compensation plus lost goodwill. This is *not* a refund, because the passenger still flies on a later flight.

With fixed fares and refunds for no-shows, this is the same as maximising expected profit per flight. The benchmark to beat is the current policy (420 / 120), which the predecessor costed at **£6,250** across both classes. That figure came from judgement-based no-show rates and has not been checked against data.

Alternative objectives worth considering

1. **Risk-adjusted cost, not just the mean.** A policy can look cheap on average and still leave a fat tail of bad flights. Measures to watch: P(any bumping), expected number of passengers bumped, and tail cost (e.g. 95th percentile or CVaR). Mike asked for a "stress-tested" recommendation, so robustness matters as well as the expected value.
2. **Robustness to the inputs we don't trust.** The goal is a policy that stays good across plausible no-show rates and plausible bump costs, not one tuned to a single guess. Two triggers for this: a competitor's more generous bumping policy may raise our goodwill cost, and the 3–8% / 15–30% no-show ranges have never been checked against the data.
3. **Customer/service-level goals.** For example, cap the bumping rate or the business-class bumps specifically. Business is the route's most valuable segment, and goodwill loss there may be worse than £300 suggests.
4. **Load factor / revenue maximisation.** Fill seats regardless of bumping. This is useful as a contrast, but it ignores goodwill, so it isn't a sensible primary goal.

Constraints and structural assumptions

- **Capacity:** 400 economy and 100 business seats (Boeing 747); 4 flights/day.

- **Booking limits:** whole numbers, at least capacity (overbooking only), set per class.
- **Pricing is fixed:** one fare per class (£450 / £1,450). Price is not a decision lever here.
- **Refund policy:** no-shows are refunded under the current policy. Finance's "no refund" option is a separate scenario.
- **Demand assumption (to check):** the standard model assumes bookings always reach the limit, i.e. demand is not binding. If demand sometimes falls short, a higher limit costs less than the model says.
- **Classes may not be independent:** gate agents informally move bumped economy passengers into spare business seats. That makes a business no-show partly offset an economy bump, which lowers the true cost of overbooking economy. It is not official policy, so we need to decide whether to model it or treat it as upside.
- **Uncertain inputs:**
  - *Aleatoric:* no-shows on a given flight vary randomly even when the true rate is known.
  - *Epistemic:* we don't know the true no-show rates (judgement ranges vs. 4 months of boarding-log data that still needs validating), or the true bump cost, which is a rough estimate and may shift with the competitor's policy.
- **Must be actionable and transferable:** Mike needs a simple booking limit he can put into practice, and wants an approach he can reuse on the other routes.

Step 2: The moving parts (inputs, outcomes, scenarios)

Decision variables (what we control)

Variable	Current	Range to explore
Economy booking limit B_E	420	≥ 400 (e.g. 400–450)
Business booking limit B_B	120	≥ 100 (e.g. 100–150)

Fixed parameters (taken as given)

Parameter	Economy	Business
Seat capacity C	400	100
Fare p (lost per empty seat)	£450	£1,450
Flights per day	4	4

Uncertain inputs

Input	Type of uncertainty	What we have	Source
No-shows on a given flight	**Aleatoric:** random flight to flight even if the rate is known	Modelled as e.g. Binomial(B, q)	Simulation
True no-show rate q_E, q_B	**Epistemic:** we don't know it	Judgement: economy 3–8% (mode 5%), business 15–30% (mode 20%), e.g. a triangular prior. Data: 4 months of boarding logs	Case + `heathrow-jfk-boarding-log.csv`
Bump cost b_E, b_B	**Epistemic:** a rough estimate, may shift	£150 / £300 base case. A competitor's more generous policy may raise our goodwill cost	Case
Bookings actually reaching the limit	**Epistemic:** assumed, not checked	Check `*_booked` against the limits in the log	Boarding log
Upgrade of bumped economy passengers into spare business seats	**Epistemic:** informal practice, extent unknown	Anecdotal only	Case

Mapping the log to the model: no-show rate ≈ 1 − `*_boarded` / `*_booked`. **Caveat:** despite its name, `*_boarded` exceeds seat capacity on more than half the flights (146/281 economy, 93/281 business). So it looks like it records **show-ups at the gate**, not passengers who actually boarded. That definition needs confirming with ops before the column is used. If it really is show-ups, it is exactly what we need. If some rows are capped at capacity, those

flights under-count show-ups and would bias the estimated no-show rate upwards. The log gets a full check in the next step.

Outcomes (per flight, per class, then summed)

- **Show-ups:** $S = B - \text{no-shows}$
- **Empty seats:** $\max(C - S, 0)$, **costing** $p\times$ empty seats
- **Bumped passengers:** $\max(S - C, 0)$, **costing** $b\times$ bumped passengers
- **Total cost** = empty-seat cost + bumping cost, across both classes. This is the headline outcome.

Summary measures across simulated flights:

- Expected total cost (main objective; compare with the £6,250 baseline)
- Spread and tail: standard deviation, 5th/95th percentile, CVaR
- P(at least one bump), expected number bumped, and the business-class bump rate
- Load factor, as a sanity check
- *No-refund scenario only:* retained revenue from no-shows, which changes profit rather than cost

Scenarios

#	Scenario	What changes
S0	**Base case / current policy**	B = 420 / 120; judgement no-show ranges; bump cost £150 / £300
S1	**Data-driven no-show rates**	Replace the judgement priors with rates estimated from the cleaned boarding log, including uncertainty in the estimate
S2	**Competitor bumping policy**	Raise the bump cost, e.g. economy £150 → £300+ and business £300 → £600+. Find the break-even point where the optimal limit changes
S3	**Upgrade practice formalised**	A bumped economy passenger takes a spare business seat when one is free, so the classes become linked
S4	**No refunds for no-shows** (finance proposal)	No-show fares are kept; possibly fewer no-shows or lower demand as a behavioural response
S5	**Stress tests**	No-show rates at the extremes of their ranges (or beyond), and demand that doesn't fill the booking limit

Step 3: Checking the boarding log before trusting it

Mike pulled `heathrow-jfk-boarding-log.csv` from ops reporting himself. Before using any of it, we load it **as is** and print every anomaly we can find. **Nothing is cleaned or changed in this step.** What to do about each issue is a separate decision, made once we've seen them all.

```
281 rows, 8 columns

flight_id           int64
date                  str
flight_number         str
economy_booked      int64
economy_boarded     int64
business_booked     int64
business_boarded    int64
notes                 str
dtype: object
```

	flight_id	date	flight_number	economy_booked	economy_boarded	business_booked	business_boarded	notes
0	1	2025-11-01	TA101	420	424	120	101	NaN
1	2	2025-11-01	TA102	420	408	120	86	NaN
2	3	2025-11-01	TA103	420	408	120	88	NaN
3	4	2025-11-01	TA104	420	398	120	51	NaN
4	5	2025-11-02	TA101	420	398	120	74	NaN

Date range and flights per day

The case says 4 flights a day (Heathrow → JFK).

```
Unparseable dates: 0
First date: 2025-11-01   Last date: 2026-01-09
Days in range: 70   Days with at least one flight: 70
Days with no flights logged: 0

Number of days by flights logged that day:
4    69
5     1

Days with a flight count other than 4:
date
2026-01-02    5

Flights logged per flight number:
flight_number
TA101    70
TA102    70
TA103    71
TA104    70
```

Duplicate rows and IDs

```
Fully identical rows (all copies shown): 2
     flight_id        date flight_number  economy_booked  economy_boarded  business_booked  business_boarded notes
250        251  2026-01-02         TA103             420              410              120               101   NaN
280        251  2026-01-02         TA103             420              410              120               101   NaN

Rows sharing a flight_id: 2
     flight_id        date flight_number  economy_booked  economy_boarded  business_booked  business_boarded notes
250        251  2026-01-02         TA103             420              410              120               101   NaN
280        251  2026-01-02         TA103             420              410              120               101   NaN

Rows sharing the same date + flight_number: 2
     flight_id        date flight_number  economy_booked  economy_boarded  business_booked  business_boarded notes
250        251  2026-01-02         TA103             420              410              120               101   NaN
280        251  2026-01-02         TA103             420              410              120               101   NaN

Rows identical except for flight_id: 2
     flight_id        date flight_number  economy_booked  economy_boarded  business_booked  business_boarded notes
250        251  2026-01-02         TA103             420              410              120               101   NaN
280        251  2026-01-02         TA103             420              410              120               101   NaN

flight_id runs from 1 to 280; 280 distinct values for 281 rows
```

Missing values and the notes column

```
Missing values per column:
flight_id             0
date                  0
flight_number         0
economy_booked        0
economy_boarded       0
business_booked       0
business_boarded      0
notes               277

Notes column, every distinct value:
notes
NaN                                                                  277
diverted (weather) -- most booked passengers rebooked onto a later flight      4

Rows with a note: 4
     flight_id        date flight_number  economy_booked  economy_boarded  business_booked  business_boarded
notes
75          76  2025-11-19         TA104             420                4              120                 6  diverted (weather) -- most booked passengers rebooked onto a later
flight
100        101  2025-11-26         TA101             420               27              120                 7  diverted (weather) -- most booked passengers rebooked onto a later
flight
176        177  2025-12-15         TA101             420               37              120                 7  diverted (weather) -- most booked passengers rebooked onto a later
flight
211        212  2025-12-23         TA104             420               22              120                 5  diverted (weather) -- most booked passengers rebooked onto a later
flight
```

Value counts of the booked columns

Under the current policy, bookings should sit at the limits: 420 economy and 120 business.

```
economy_booked:
economy_booked
420    281

business_booked:
business_booked
120    281
```

Boarded vs. booked vs. capacity, per class

Physically, more passengers can't board than there are seats (400 economy / 100 business). They also can't board without a booking. A pile-up of values just at or around capacity would suggest the column is capped or recorded differently on full flights.

```
=== ECONOMY (capacity 400) ===
Non-numeric booked / boarded values: 0 / 0
boarded > booked:    3
boarded > capacity:  146
boarded = capacity:  11
boarded range:       4 to 425
```

```
Flights at each boarded count from 397 to 403:
economy_boarded
397     6
398    15
399     5
400    11
401    10
402     8
403     8
Rows where boarded > booked:
     flight_id        date flight_number  economy_booked  economy_boarded  business_booked  business_boarded notes
0            1  2025-11-01         TA101             420              424              120               101   NaN
96          97  2025-11-25         TA101             420              425              120                72   NaN
230        231  2025-12-28         TA103             420              424              120                79   NaN

=== BUSINESS (capacity 100) ===
Non-numeric booked / boarded values: 0 / 0
boarded > booked:    0
boarded > capacity:  93
boarded = capacity:  6
boarded range:       5 to 118
Flights at each boarded count from 97 to 103:
business_boarded
97      5
98      9
99      6
100     6
101    14
102    10
103    10
```

Step 3b: Cleaning the log and describing no-shows

Cleaning decisions:

- **Drop the duplicate** of flight 251. It is the same flight logged twice.
- **Drop the 4 weather diversions.** Passengers were rebooked by the airline, so those flights tell us nothing about no-show behaviour.
- **Keep the 3 economy flights where boarded > booked.** That shouldn't be possible, but we have no basis to correct them. They give slightly negative no-show rates, and they are flagged below so they stay visible.

No-show rate per flight = 1 − boarded / booked. This assumes `*_boarded` records **show-ups at the gate**. Step 3 suggests that is what it means, but ops still needs to confirm it.

```
Rows in raw log:                 281
After dropping exact duplicates:  280  (removed 1)
After dropping diverted flights:  276  (removed 4)

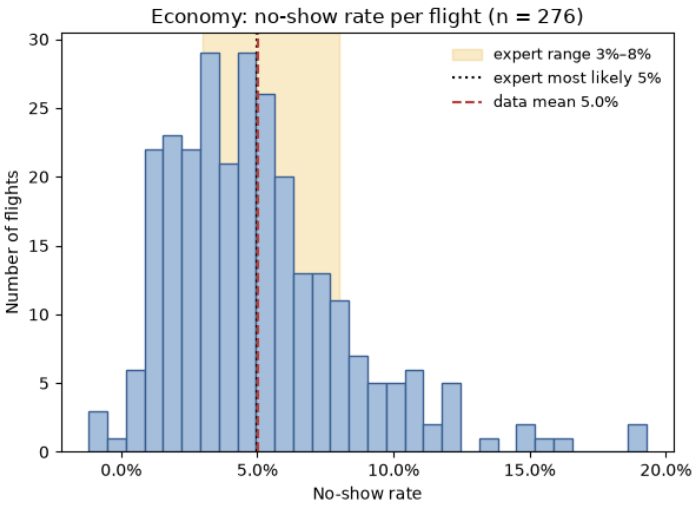
Flights remaining for analysis: 276

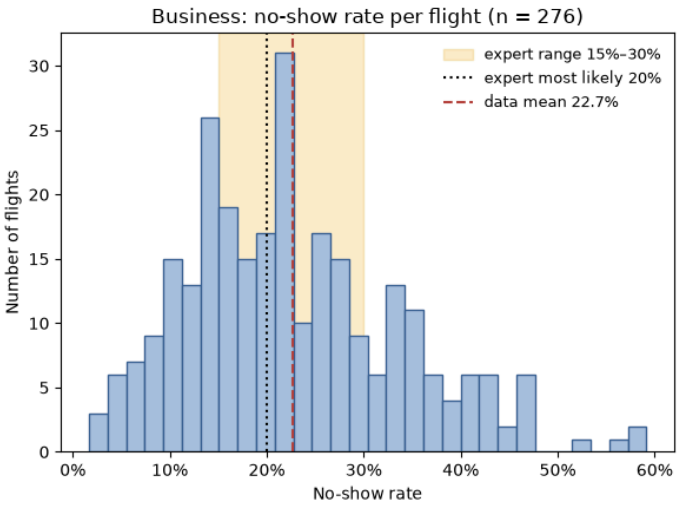
Flights with a negative no-show rate (boarded > booked), kept but flagged: 3
     flight_id       date flight_number  economy_booked  economy_boarded  econ_noshow_rate
0            1 2025-11-01         TA101             420              424         -0.009524
96          97 2025-11-25         TA101             420              425         -0.011905
230        231 2025-12-28         TA103             420              424         -0.009524
```

class	econ_noshow_rate	bus_noshow_rate
flights	276.000	276.000
mean	0.050	0.227
median	0.045	0.208
std	0.033	0.112
min	-0.012	0.017
p5	0.010	0.067
p95	0.111	0.433
max	0.193	0.592
expert_low	0.030	0.150
expert_mode	0.050	0.200
expert_high	0.080	0.300
share_inside_expert_range	0.558	0.500
share_below_expert_range	0.279	0.250
share_above_expert_range	0.163	0.250

Distribution of no-show rates vs. the expert range

The shaded band is the experts' 3–8% (economy) and 15–30% (business) range. The dotted line is their most likely value, and the dashed line is the mean in the data.

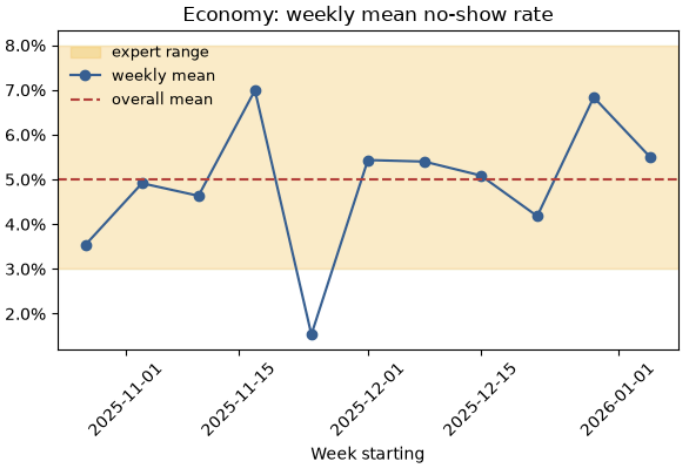


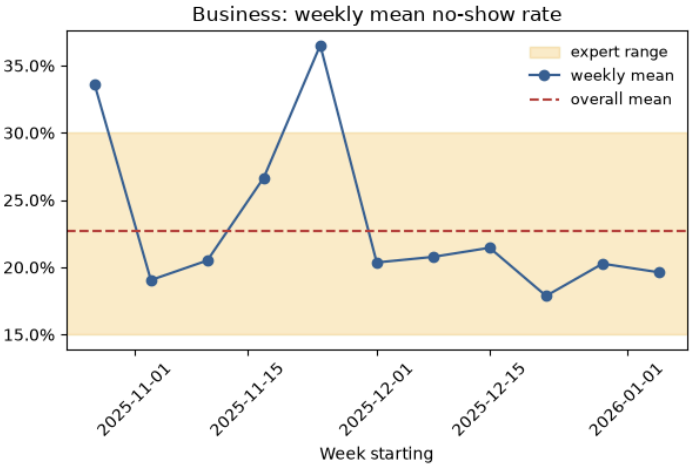


Weekly trend

Is the no-show rate stable over the 10 weeks, or drifting? Weeks run Monday to Sunday. The first and last weeks may be partial, so the number of flights per week is shown too.

week	flights	econ_mean	bus_mean
2025-10-27	8	0.035	0.336
2025-11-03	28	0.049	0.190
2025-11-10	28	0.046	0.205
2025-11-17	27	0.070	0.267
2025-11-24	27	0.015	0.365
2025-12-01	28	0.054	0.204
2025-12-08	28	0.054	0.208
2025-12-15	27	0.051	0.215
2025-12-22	27	0.042	0.179
2025-12-29	28	0.068	0.203
2026-01-05	20	0.055	0.196





Means by weekday, weekend vs. weekday, and flight number

Groups are small (about 40 flights per weekday and 70 per flight number), so differences of a point or so may be noise. The standard deviation and count are shown to help judge that.

weekday	flights	econ_mean	econ_std	bus_mean	bus_std
Monday	39	0.053	0.031	0.193	0.104
Tuesday	39	0.038	0.025	0.213	0.089
Wednesday	38	0.056	0.045	0.233	0.124
Thursday	40	0.050	0.036	0.175	0.118
Friday	40	0.043	0.021	0.221	0.106
Saturday	40	0.054	0.031	0.268	0.109
Sunday	40	0.057	0.036	0.283	0.094

is_weekend	flights	econ_mean	econ_std	bus_mean	bus_std
weekday (Mon–Fri)	196	0.048	0.033	0.207	0.110
weekend (Sat–Sun)	80	0.055	0.033	0.276	0.101

flight_number	flights	econ_mean	econ_std	bus_mean	bus_std
TA101	68	0.049	0.031	0.230	0.112
TA102	70	0.052	0.031	0.221	0.109
TA103	70	0.054	0.038	0.228	0.116
TA104	68	0.046	0.031	0.228	0.111

Correlation between the classes

Do economy and business no-shows move together on the same flight? If they do, the model should not draw them independently.

```
Correlation between economy and business no-show rates on the same flight (n = 276):
  Pearson:  0.065
  Spearman: -0.014
```

What the near-zero correlation hides. Over all 276 flights the correlation is about 0, but that comes from two outlier periods pulling in opposite directions:

- **Thanksgiving week (24–30 Nov)** pushes the cabins *apart*: economy no-shows almost disappear (~1.5%) while business jumps (~36%).
- **19–20 Nov**, around the weather diversion, pushes them *together*: both cabins are high at once (economy 12–14%, business 40–45%).

Sample (checked outside this notebook)	Pearson
All flights	0.07
Excluding Thanksgiving week	0.25
Excluding 19–20 Nov	−0.07
Excluding both	0.12 (p ≈ 0.07)

So the accurate reading is **weakly positive on normal days, with disruption days moving both cabins together**. It is not "independent". Disruption days are exactly the ones that drive bumping risk, so the simulation should be run with independent cabins *and* with a mild positive correlation, as a sensitivity check.

Step 4: How variable are no-shows, and what distribution should the model use?

Step 3b gave the average no-show rate. The simulation also needs to know how much the rate **varies from flight to flight**, and how precisely we know the average.

- **Dispersion ratio.** If every passenger no-shows independently with the same probability q, the per-flight rate has binomial variance $q(1-q)/n$ (n = 420 or 120 booked). The ratio *observed variance ÷ binomial variance* is about 1 if a plain Binomial(B, q) is enough. Well

above 1 means the rate itself shifts from flight to flight, and a binomial model would understate bumping risk. The p-value tests ratio = 1 ($(k-1)\times$ ratio $\sim \chi^2_{k-1}$ for k flights).

- **Beta fit (method of moments)** for the *underlying* per-flight rate p, so the simulation can draw $p \sim$ Beta(α, β) and then no-shows $\sim$ Binomial(B, p). The observed rate already contains binomial noise, so that part is removed first: $\mathrm{Var}(p) = \big(\text{observed variance} - m(1-m)/n\big)/(1-1/n)$. Then $\alpha+\beta = m(1-m)/\mathrm{Var}(p) - 1$, with $\alpha = m(\alpha+\beta)$.
- **Bootstrap 95% CI for the mean rate**, 10,000 resamples. Resampling individual flights treats them as independent. Step 3b showed that flights on the same day move together, so a second CI resamples **whole days**. That is the more honest one.
- **Ordinary days** exclude the two disruption periods found in Step 3b: 19–20 Nov (weather) and Thanksgiving week (24–30 Nov). That leaves the pattern on a typical day; the all-flights rows keep the disruptions in.
- **Business by weekday vs. weekend**, because Step 3b showed business no-shows are clearly higher on Saturdays and Sundays.

The expert triangular distributions (3%, 5%, 8%) and (15%, 20%, 30%) sit in the same table for comparison: mean, SD and 5th/95th percentiles.

```
All flights:          276 flights on 70 days
Ordinary days only:   242 flights on 61 days
Excluded as disrupted: 34 flights on 9 days
```

class	sample	flights	mean	CI low (flights)	CI high (flights)	CI low (days)	CI high (days)	observed SD	binomial SD	dispersion ratio	p-value (ratio = 1)	beta alpha	beta beta	SD	p5	p9
economy	expert triangular (3%, 5%, 8%)		5.3%											1.0%	3.7%	7.1
	all flights	276	5.0%	4.6%	5.4%	4.5%	5.6%	3.3%	1.1%	9.7	< 0.001	2.4	44.9	3.1%	1.1%	11.0
	ordinary days	242	5.2%	4.8%	5.5%	4.7%	5.6%	2.9%	1.1%	7.3	< 0.001	3.4	62.2	2.7%	1.6%	10.3
business	expert triangular (15%, 20%, 30%)		21.7%											3.1%	16.9%	27.3
	all flights	276	22.7%	21.4%	24.0%	20.8%	24.6%	11.2%	3.8%	8.5	< 0.001	3.4	11.5	10.5%	7.7%	42.0
	ordinary days	242	20.6%	19.3%	21.8%	19.0%	22.1%	9.7%	3.7%	6.9	< 0.001	3.9	15.1	9.0%	7.6%	37.1
	weekday (Mon–Fri), all	196	20.7%	19.2%	22.2%	18.5%	22.9%	11.0%	3.7%	8.8	< 0.001	2.9	11.3	10.4%	6.2%	39.9
	weekend (Sat–Sun), all	80	27.6%	25.4%	29.8%	25.0%	30.4%	10.1%	4.1%	6.1	< 0.001	6.1	16.0	9.3%	13.5%	44.0
	weekday (Mon–Fri), ordinary	170	18.2%	16.9%	19.6%	16.7%	19.8%	8.9%	3.5%	6.4	< 0.001	3.8	17.2	8.2%	6.6%	33.3
	weekend (Sat–Sun), ordinary	72	26.1%	24.1%	28.3%	24.1%	28.2%	9.3%	4.0%	5.4	< 0.001	6.8	19.3	8.4%	13.3%	41.0

Reading the table. *SD*, *p5* and *p95* describe the per-flight rate: from the fitted beta for the data rows, and from the triangular distribution for the expert rows. *Observed SD* is the raw spread of the logged rates, which includes binomial noise.

What it shows

- **The means agree with the experts.** Economy is about 5.0–5.2% against an expert mean of 5.3%. Business is 20.6% on ordinary days (22.7% with the disruptions) against 21.7%. Even the day-level CIs are narrow, about ±0.5 points for economy and ±1.5–2 points for business. The *average* is not the big uncertainty.
- **A plain binomial is wrong for both classes.** The observed variance is 7–10 times the binomial variance, and still 5–7 times on ordinary days, so the excess is not just the disruptions. The rate itself moves from flight to flight. Binomial(B, q) with a fixed q would make bumping look much rarer than it is.
- **Per-flight spread is about three times wider than the expert triangle.** On ordinary days the beta fit gives economy 1.6–10.3% (5th–95th percentile) against the experts' 3.7–7.1%, and business 7.6–37% against 17–27%. The experts' ranges are a fair guess at the *mean* rate. They are far too narrow as a *per-flight* distribution.
- **Business weekends are a different population.** The ordinary-day mean is 18.2% on weekdays and 26.1% on weekends, and the day-level CIs (16.7–19.8% vs. 24.1–28.2%) don't overlap. A single business limit would be too tight on weekends or too loose on weekdays, so a day-of-week limit is worth testing.

For the simulation: draw the per-flight rate from the ordinary-day beta, e.g. economy Beta(3.4, 62.2) and business Beta(3.8, 17.2) weekdays / Beta(6.8, 19.3) weekends, then draw no-shows ~ Binomial(B, rate). Disruption days need to be handled separately (a scenario, or a mixture with their frequency of about 9 days in 70) rather than averaged in.

Step 5: Reproduce the £6,250, then build the simulation

5a. The predecessor's deterministic cost

The predecessor put the most likely no-show rates (5% economy, 20% business) into the booking limits and costed the **expected** show-ups as if they were certain:

- Economy: 420 × 95% = 399 show up, leaving 1 empty seat × £450 = £450
- Business: 120 × 80% = 96 show up, leaving 4 empty seats × £1,450 = £5,800

No one is bumped in either class, so the total is £6,250. The model function below is the cost model used for the rest of the notebook, with the deterministic show-ups plugged in.

```
Predecessor's deterministic cost per flight: £6,250
```

Run result

Objective: `total_cost` = 6,250.00

Outcomes:

economy_shows	399.00
economy_empty	1.00
economy_bumped	0.00
economy_cost	450.00
business_shows	96.00
business_empty	4.00
business_bumped	0.00
business_cost	5,800.00
total_cost	6,250.00

5b. Two-level Monte Carlo

Outer loop (epistemic): what is the true mean no-show rate? 1,000 draws of the pair (q_E, q_B) from either:

- **Expert:** the triangular distributions (3%, 5%, 8%) and (15%, 20%, 30%), drawn independently, or
- **Data:** the day-level bootstrap of the ordinary-day mean from Step 4. Each draw resamples the 61 ordinary days and takes the mean over their flights, which keeps any link between the two cabins' means.

Inner loop (aleatoric): what happens on each flight? 2,000 flights per outer draw (500 days × 4 flights). Each flight's no-show rate is drawn from Beta with mean q and the **ordinary-day concentration** $\alpha + \beta$ fitted in Step 4. Show-ups are then Binomial(B, 1 − rate), which is the beta-binomial. **Assumption:** the expert inputs use the same flight-to-flight spread as the data, because the experts only gave a range for the mean. This way the two input sources differ only in the mean.

Correlation between cabins is set through a day-level factor shared by the 4 flights of a day. It enters a Gaussian copula, so each cabin's beta marginal stays exactly as above:

Setting	Economy rate	Business rate	Cabin correlation (latent)
Independent (base case)	own noise only	own noise only	0
Common shock	+ day factor	+ day factor	+0.4
Divergent (Thanksgiving-type)	− day factor	+ day factor	−0.4

The ±0.4 is a stress assumption, not an estimate. The flight-level correlation was 0.12–0.25 on normal days in Step 3b, and disruption days push it further in either direction.

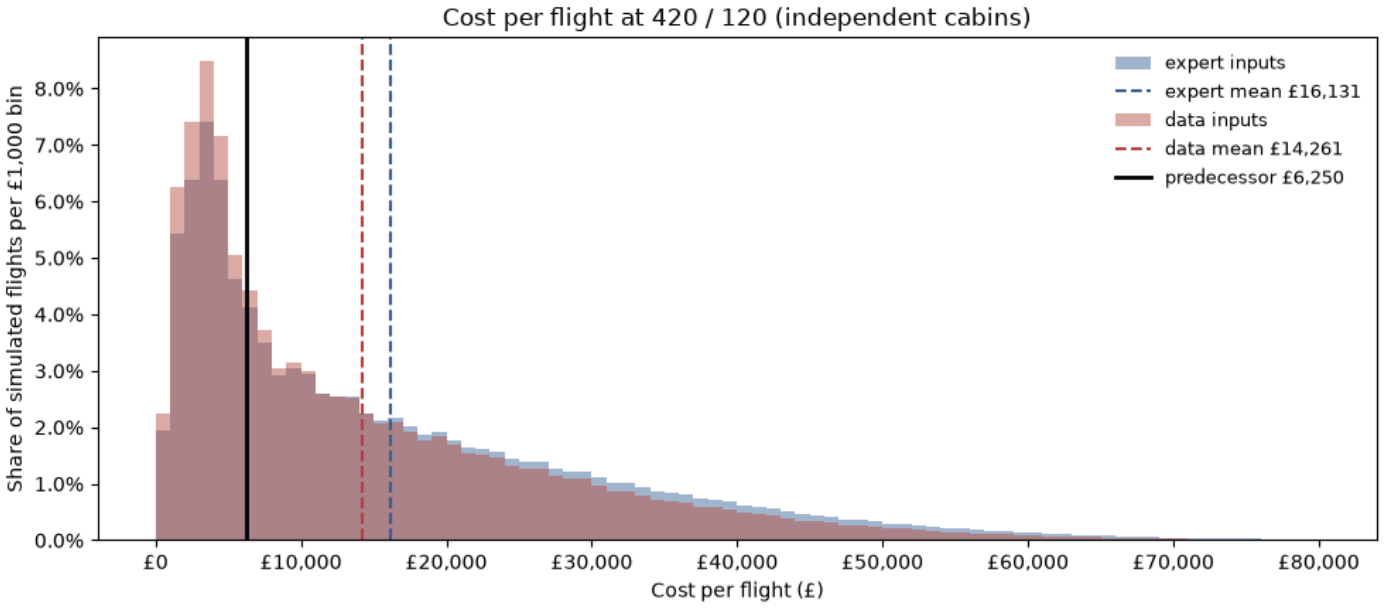
Common random numbers. Every random number (outer draws, day factors, per-flight noise, show-up uniforms) is drawn once and reused for every booking limit, input source and correlation setting. Show-ups for the first C bookings (400 / 100) come from the binomial inverse CDF of a fixed uniform. Each booking above capacity then gets its own fixed coin flip: booking number 421 is the same simulated passenger, who shows or doesn't, at every limit from 421 upwards. So a higher limit never gives fewer show-ups on the same simulated flight, and a sweep over limits (Step 6) only adds one coin flip per extra booking. Differences between policies are then due to the policy, not to simulation noise.

Beta concentration (alpha + beta), ordinary days: economy 65.6, business 19.0

inputs	correlation	mean cost	SD	P95	CVaR95	P(any bump)	P(business bump)	expected bumps, economy	expected bumps, business	90% band low (mean cost)	90% band high (mean cost)
expert	independent	£16,131	£14,551	£45,700	£55,684	64.8%	33.9%	3.77	2.31	£10,895	£23,279
	common shock	£16,141	£15,457	£47,800	£58,841	60.2%	33.9%	3.76	2.31	£10,844	£23,349
	divergent	£16,139	£13,834	£44,100	£53,658	69.3%	33.9%	3.77	2.31	£10,913	£23,137
data	independent	£14,261	£13,276	£41,550	£51,147	68.2%	36.9%	3.91	2.50	£12,682	£15,816
	common shock	£14,264	£14,152	£43,650	£54,218	62.9%	36.9%	3.91	2.50	£12,606	£15,999
	divergent	£14,265	£12,639	£40,050	£49,254	73.4%	36.9%	3.91	2.50	£12,752	£15,909

Cost distribution under the current policy (420 / 120)

Independent cabins, expert vs. data inputs. Every simulated flight from every outer draw is included. The black line is the predecessor's £6,250, and the dashed lines are the simulated means.



Check: what did the logged flights actually cost?

Every logged flight was flown at 420 / 120, and `*_boarded` is taken as show-ups (Step 3), so the actual cost of each flight can be computed directly. The **data / independent** simulation should land close to the ordinary-day figure.

sample	flights	mean cost	SD	P95	P(any bump)	P(business bump)
actual, all flights	276	£17,415	£16,945	£49,338	68.1%	33.3%
actual, ordinary days	242	£14,053	£13,591	£40,050	66.5%	38.0%
simulated, data / independent	2,000,000	£14,261	£13,276	£41,550	68.2%	36.9%

What Step 5 shows

- **The £6,250 is reproduced, and it is far too low.** The real expected cost at 420 / 120 is about **£14,300** with data inputs and **£16,100** with expert inputs, 2.3–2.6 times the predecessor's figure. The log confirms this: ordinary-day flights actually cost **£14,053** on average, and £17,415 including disruption days.
- **Why:** costing the *average* show-up hides the asymmetry. An empty business seat costs £1,450 and a business bump only £300, so the flights where more people than usual no-show cost far more than the flights where fewer do save. With the per-flight spread found in Step 4, business empty seats alone are worth about £10,000 per flight.
- **Bumping is routine, not rare.** About two-thirds of flights bump someone, and 34–37% bump at least one business passenger. On average that is about 3.8–3.9 economy and 2.3–2.5 business passengers bumped per flight. The deterministic model said zero.
- **The tail is heavy.** P95 is about £42–46k and CVaR95 about £51–56k per flight.
- **Expert vs. data:** the means are close, but the expert triangle leaves far more doubt about the true mean. The 90% band on expected cost is **£10.9k–£23.3k** for expert inputs and **£12.7k–£15.8k** for data inputs.

- **Correlation does not move the mean** (at £150 / £300 with no upgrades, the cabins are costed separately, so expected cost is additive). It moves the risk. A common shock raises the SD and P95 by about £1–2k and lowers P(any bump), because bad flights in the two cabins tend to fall on the same days. A divergent pattern does the opposite. This will matter once upgrades link the cabins (scenario S3).

Next: sweep the booking limits under these same random numbers to find the cost-minimising 420 / 120 replacement, and check how robust it is across the input sources and correlation settings.

Step 6: Optimise the booking limits

Grid search over economy 400–460 × business 100–160 (61 × 61 = 3,721 policies), on the **same simulated flights** as Step 5, with independent cabins, for expert and for data inputs.

How the grid is computed. With no upgrades between cabins (the base case), a flight's cost is economy cost + business cost, and each depends only on its own limit. So each cabin's cost is simulated once per limit (61 + 61 runs), and the expected cost of every grid cell is the sum of the two. That gives exactly the same answer as simulating all 3,721 pairs. It also means the best economy limit does not depend on the business limit, and vice versa. If upgrades are modelled later (S3), this shortcut no longer holds and the grid has to be simulated jointly.

Epistemic bands. Each outer draw (a possible true mean no-show rate) gives its own expected cost curve. The band is the 5th–95th percentile of those curves. Because every limit uses the same simulated flights, the *saving* of one policy over another is computed draw by draw. Its band and P(saving > 0) are much tighter than the overlap of two cost bands would suggest.

```
EXPERT inputs
  Best limits:     economy 430, business 142   expected cost £9,983 per flight
  Current 420/120: expected cost £16,131 per flight

DATA inputs
  Best limits:     economy 429, business 139   expected cost £9,223 per flight
  Current 420/120: expected cost £14,261 per flight
```

How flat is the bottom?

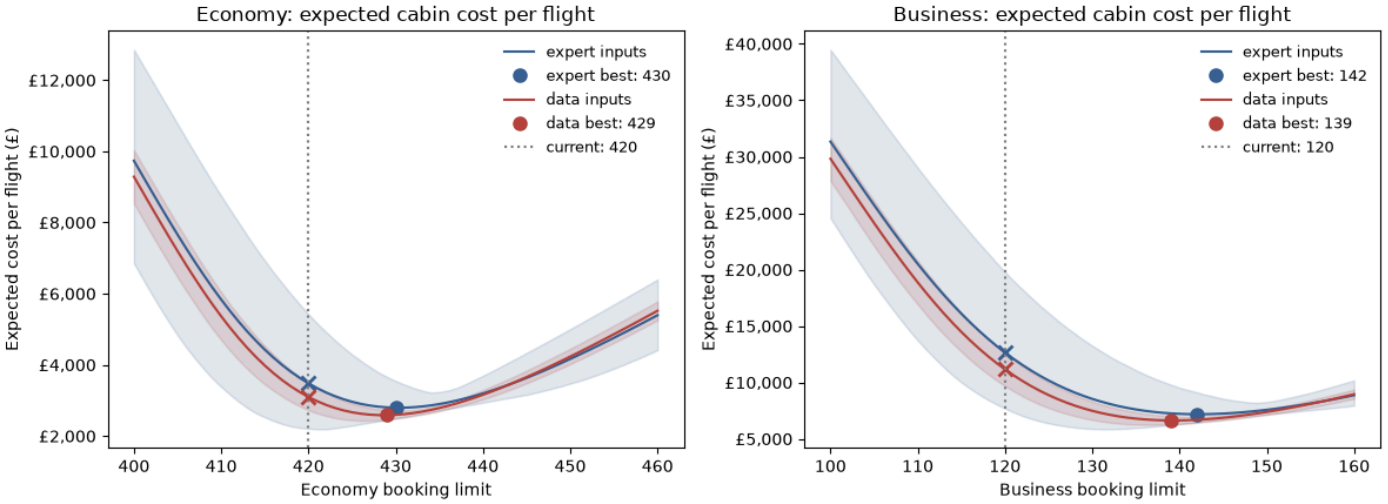
The cost curves are flat near the minimum. The limits below keep each cabin's expected cost within £100 per flight (£146k a year) of its best. Choosing the **low end** of a range gives up almost nothing in expected cost and cuts bumping.

inputs	cabin	best limit	near-optimal from	near-optimal to	best cabin cost
expert	economy	430	427	434	£2,796
	business	142	139	145	£7,187
data	economy	429	425	433	£2,587
	business	139	136	142	£6,636

Cost curves with epistemic bands

Each panel shows one cabin's expected cost per flight (the other cabin's cost doesn't depend on this limit). Lines are the expected cost averaged over the outer draws. Shaded bands are

the 5th–95th percentile across the outer draws, i.e. how much the curve moves with the true mean no-show rate. Dots mark the best limit and crosses the current one.



Best vs. current

Each policy is evaluated under both input sources, so the table also shows whether the limits chosen under one set of inputs still save money under the other. *Saving* is against 420 / 120 under the same inputs. It is computed per outer draw on the same flights: the 90% band is the 5th–95th percentile of the annual saving across draws, and P(saving > 0) is the share of draws where the policy beats 420 / 120. Annual figures are per flight × 4 flights × 365 days.

inputs	policy	cost per flight	annual cost	annual saving	saving 90% low	saving 90% high	P(saving > 0)	P95 per flight	P(any bump)	P(business bump)	bumps per flight
expert	current 420 / 120	£16,131	£23,551,234	£0	£0	£0	0.0%	£45,700	64.8%	33.9%	6.08
	best under expert 430 / 142	£9,983	£14,575,048	£8,976,186	£2,271,165	£16,555,767	99.7%	£25,850	93.9%	78.0%	23.20
	best under data 429 / 139	£10,067	£14,697,187	£8,854,047	£2,814,196	£15,523,957	100.0%	£28,200	92.3%	74.1%	20.60
data	current 420 / 120	£14,261	£20,820,755	£0	£0	£0	0.0%	£41,550	68.2%	36.9%	6.41
	best under expert 430 / 142	£9,291	£13,564,481	£7,256,273	£5,146,681	£9,283,529	100.0%	£21,900	95.6%	81.8%	24.62
	best under data 429 / 139	£9,223	£13,466,065	£7,354,690	£5,470,169	£9,191,669	100.0%	£23,950	94.2%	78.1%	21.88

Separate weekday and weekend business limits

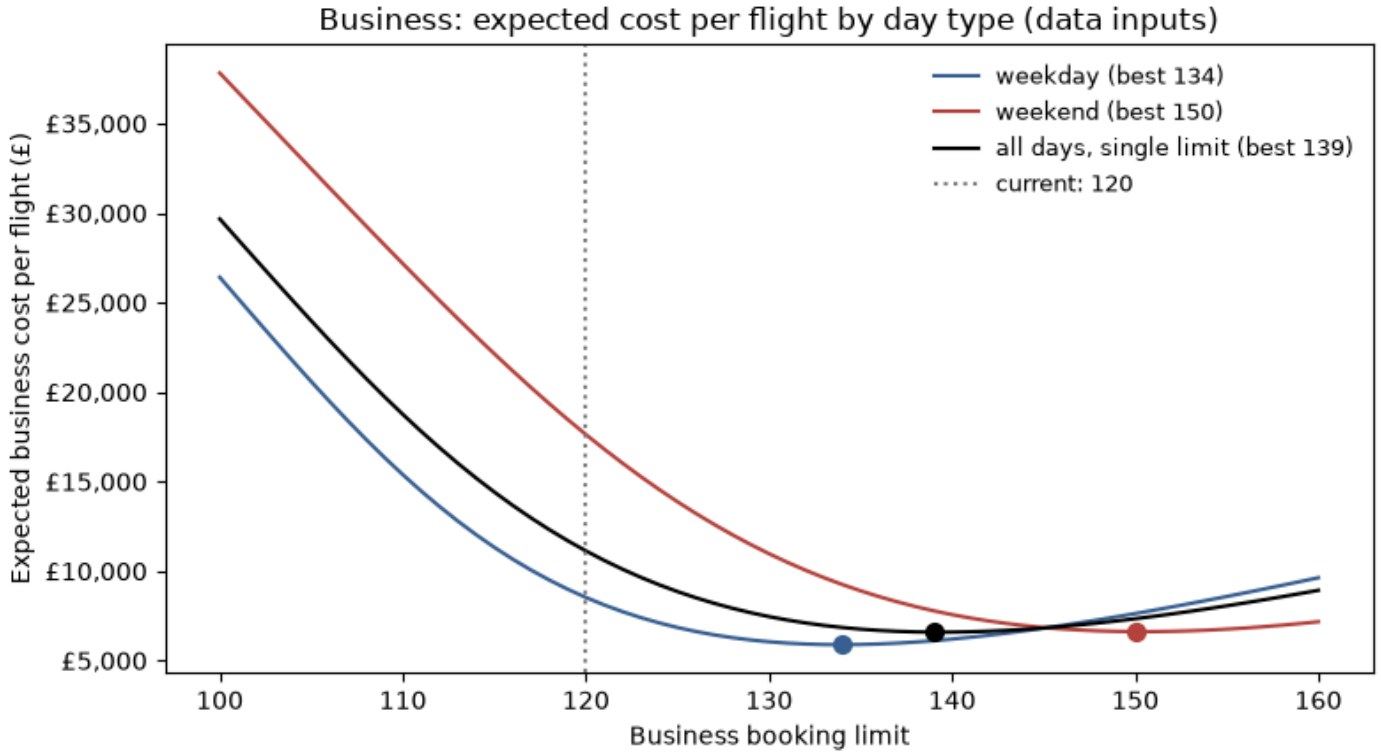
Step 4 found that the business no-show rate is clearly higher at weekends (ordinary days: 18.2% weekdays vs. 26.1% weekends). This test uses **data inputs only**, since the experts gave no weekday/weekend split:

- Outer draws bootstrap the weekday and weekend ordinary days separately, giving a weekday and a weekend mean for each draw.
- Each day type uses its own ordinary-day beta concentration from Step 4.
- The same simulated flights are run once as weekday flights and once as weekend flights. Weekdays are weighted 5/7 and weekends 2/7.

The comparison is **split limits vs. the best single business limit under this same model**, so the gain is due only to splitting.

```
Annual gain of splitting over the best single limit: £717,457  (90% band £308,721 to £1,148,546, P(gain > 0) = 100.0%)
As a share of the annual business cost under the single best limit: 7.5%
```

business policy	business cost per flight	annual business cost
current 120 every day	£11,125	£16,242,065
single best 139 every day	£6,588	£9,618,562
split: 134 weekdays / 150 weekends	£6,097	£8,901,105



What Step 6 shows

- **Best limits: economy 429–430, business 139–142.** Data inputs give **429 / 139** and expert inputs **430 / 142**. Both are well above the current 420 / 120, and neither is at the edge of the grid.
- **The choice is robust to which inputs you trust.** Each set of limits is within about £85 per flight of the other's optimum, whichever inputs are used to judge it. The flat bottom is wide: economy 425–434 and business 136–145 are all within £100 per flight of the best.
- **The saving is large and almost certain** at the base-case costs. With data inputs, the annual cost falls from **£20.8M to £13.5M**, a saving of **£7.4M a year** (90% band £5.5M–£9.2M, P(saving > 0) = 100%). With expert inputs the saving is £9.0M (90% band £2.3M–£16.6M, P = 99.7%). The wider band reflects the experts' wider range for the mean.
- **But it buys that saving with a lot of bumping.** About 22 passengers are bumped per flight (against about 6 today), and 78% of flights bump at least one business passenger (against 37%). This is what the model *should* do while a bump costs £150 / £300 against fares of £450 / £1,450, so it is only as good as those bump costs. At this scale, goodwill costs, the competitor's policy (S2) and a service-level cap on business bumps all need checking before any of these limits is recommended. It also assumes bookings fill to 139 business. The log can't show that, because bookings always hit the current 120 limit.

Weekday / weekend business limits (data inputs). The best split is **134 on weekdays and 150 at weekends**, against a best single limit of 139. It saves **£0.72M a year** over that single limit (90% band £0.31M–£1.15M, P(gain > 0) = 100%), which is 7.5% of the business cost. That is about a tenth of what re-optimising the single limits already gains.

Is it worth the complexity? On the numbers, yes. The gain is clearly positive in every outer draw, and the extra complexity is one more number in the booking system, which most airline systems already set by day of week. Two cautions:

- **Thin weekend data.** The weekend estimate rests on 72 flights (18 ordinary weekend days).
- **More weekend bumping.** The weekend limit of 150 makes the bumping problem above worse on exactly the days with the most business travellers.

So: adopt the single limits first, and add the weekend split once the bump-cost and service-level checks confirm that heavy business overbooking is acceptable.

Step 7: Stress tests

Step 6 found limits of about **429 / 139** that save £7–9M a year at the base-case costs. This step checks how far that holds up:

- **7a. Bump cost:** the competitor's more generous bumping policy, as multiples of £150 / £300.
- **7b. Correlation:** the same policies with common-shock and divergent cabins, focusing on the tail.
- **7c. Upgrades:** bumped economy passengers moved into spare business seats, re-optimised jointly.
- **7d. Regret table:** five candidate policies against seven possible worlds.
- **7e. Sensitivity:** a tornado chart of which input moves the cost and the best limits most.

Everything runs on the **same simulated flights** (common random numbers) as Steps 5 and 6, so differences between policies and worlds are not simulation noise. The policy under test, called the *recommendation* below, is the Step 6 best on data inputs, 429 / 139.

7a. Bump cost: the competitor's policy

A more generous competitor raises what a bump costs us in goodwill, but by how much is unknown. So both bump costs are multiplied by 1, 1.5, 2, 3, 4 and 6 (×6 is £900 / £1,800).

One sweep covers every multiplier. A cabin's expected cost is fare × E[empty seats] + bump cost × E[bumped]. Neither expectation depends on the bump cost, so each cabin is swept once over the limits, and the cost curve for any multiplier is a weighted sum of the two. The first cell checks that multiplier 1 reproduces Step 6's curves exactly.

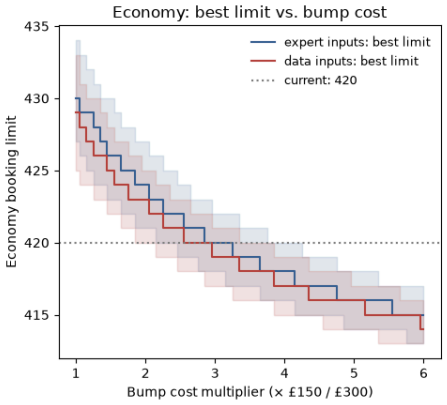
Break-even. The annual saving of a fixed policy over 420 / 120 is a straight line in the multiplier m: saving$(m) = a - m\,b$, where a is the fare revenue recovered from fewer empty

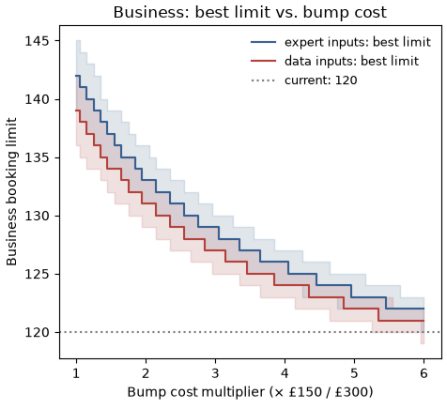
seats and b is the extra bump cost at £150 / £300. The recommendation stops beating 420 / 120 at exactly $m^* = a/b$. Solving this per outer draw gives a 90% band for m^*.

Step 6 cost curves reproduced at multiplier 1

inputs	bump cost ×	economy £	business £	best economy	economy near-optimal	best business	business near-optimal	cost at best	cost at 420 / 120	cost at 429 / 139	annual saving, 429 / 139 vs 420 / 120	P(saving > 0)	annual saving, re-optimised vs 420 / 120
expert	1.000000	£150	£300	430	427–434	142	139–145	£9,983	£16,131	£10,067	£8,854,047	100.0%	£8,976,186
	1.500000	£225	£450	426	423–430	137	134–139	£12,383	£16,760	£12,480	£6,249,930	93.2%	£6,391,000
	2.000000	£300	£600	424	421–427	133	131–136	£14,239	£17,390	£14,893	£3,645,812	74.1%	£4,599,635
	3.000000	£450	£900	420	418–422	129	127–130	£16,999	£18,648	£19,718	£-1,562,422	37.8%	£2,408,245
	4.000000	£600	£1,200	418	416–420	126	124–127	£19,004	£19,907	£24,544	£-6,770,657	17.7%	£1,318,845
	6.000000	£900	£1,800	415	413–417	122	120–123	£21,795	£22,424	£34,196	£-17,187,125	2.4%	£919,527
data	1.000000	£150	£300	429	425–433	139	136–142	£9,223	£14,261	£9,223	£7,354,690	100.0%	£7,354,690
	1.500000	£225	£450	425	422–428	134	132–137	£11,453	£14,929	£11,796	£4,574,769	100.0%	£5,075,822
	2.000000	£300	£600	423	420–425	131	129–133	£13,182	£15,598	£14,369	£1,794,847	88.9%	£3,527,770
	3.000000	£450	£900	419	417–421	127	125–129	£15,752	£16,935	£19,514	£-3,764,996	1.2%	£1,726,942
	4.000000	£600	£1,200	417	415–419	124	123–126	£17,626	£18,273	£24,659	£-9,324,839	0.0%	£943,734
	6.000000	£900	£1,800	414	413–416	121	119–122	£20,257	£20,947	£34,950	£-20,444,525	0.0%	£1,007,348

policy	judged on	break-even bump cost ×	economy bump cost £	business bump cost £	90% band low ×	90% band high ×
429 / 139 (best on data)	expert inputs	2.70	£405	£810	1.43	5.29
	data inputs	2.32	£348	£697	1.91	2.79
430 / 142 (best on expert)	expert inputs	2.45	£367	£735	1.29	4.74
	data inputs	2.10	£315	£630	1.73	2.51







What 7a shows

- **The best limits fall steadily as bumping gets dearer.** On data inputs they go from 429 / 139 (×1) to 425 / 134 (×1.5), 423 / 131 (×2), 419 / 127 (×3), 417 / 124 (×4) and 414 / 121 (×6). Expert inputs run one to three seats higher at every multiplier. **Economy drops below today's 420 from about ×3. Business stays above 120 even at ×6**, because an empty business seat (£1,450) still costs about as much as a business bump at ×6 (£1,800).
- **The recommendation stops beating 420 / 120 at about ×2.3 on data inputs**, i.e. bump costs of about **£350 economy / £700 business** (90% band ×1.9–×2.8). On expert inputs the break-even is ×2.7, with a much wider band (×1.4–×5.3). At ×2 the recommendation still saves £1.8M a year (P = 89%). At ×3 it *loses* £3.8M a year against today's policy.
- **Re-optimising always beats 420 / 120.** Even at ×6 the right limits for that bump cost save about £1.0M a year. The risk is not in overbooking more as such. It is in **locking in**

429 / 139 while the bump cost moves.

- **So the limits should be tied to the bump cost.** If the competitor's policy pushes our real goodwill cost above about twice today's estimate, the limits need to come down along the curve above. Estimating that goodwill cost properly is worth more than any further refinement of the no-show model.

7b. Correlation between the cabins

The Step 6 best policies are rerun under the **common-shock** (latent correlation +0.4) and **divergent** (−0.4, Thanksgiving-type) settings from Step 5, on the same simulated flights. The 420 / 120 rows reuse Step 5's results. Without upgrades the cabins are costed separately, so correlation cannot change the expected cost; what it can change is the **tail** and how often bumping happens. The last three columns are the change from the independent case for the same policy and inputs.

inputs	policy	correlation	mean cost	SD	P95	CVaR95	P(any bump)	P(business bump)	P95 vs independent	CVaR95 vs independent	P(any bump) vs independent
expert	current 420 / 120	independent	£16,131	£14,551	£45,700	£55,684	64.8%	33.9%	£0	£0	+0.0%
		common shock	£16,141	£15,457	£47,800	£58,841	60.2%	33.9%	£2,100	£3,157	-4.6%
		divergent	£16,139	£13,834	£44,100	£53,658	69.3%	33.9%	£-1,600	£-2,026	+4.5%
	best 430 / 142	independent	£9,983	£8,032	£25,850	£36,881	93.9%	78.0%	£0	£0	+0.0%
		common shock	£9,983	£8,309	£26,300	£38,375	90.5%	78.0%	£450	£1,493	-3.4%
		divergent	£9,982	£8,079	£26,050	£36,667	96.8%	78.0%	£200	£-214	+2.9%
data	current 420 / 120	independent	£14,261	£13,276	£41,550	£51,147	68.2%	36.9%	£0	£0	+0.0%
		common shock	£14,264	£14,152	£43,650	£54,218	62.9%	36.9%	£2,100	£3,070	-5.3%
		divergent	£14,265	£12,639	£40,050	£49,254	73.4%	36.9%	£-1,500	£-1,893	+5.2%
	best 429 / 139	independent	£9,223	£7,467	£23,950	£34,353	94.2%	78.1%	£0	£0	+0.0%
		common shock	£9,221	£7,751	£24,400	£35,804	90.4%	78.0%	£450	£1,451	-3.8%
		divergent	£9,221	£7,531	£24,150	£34,156	97.3%	78.0%	£200	£-198	+3.1%

What 7b shows

- **The mean doesn't move** (within £10 per flight), as expected without upgrades.
- **A common shock fattens the tail, but much less for the recommendation than for today's policy.** P95 rises by about £450 at 429 / 139 against £2,100 at 420 / 120, and CVaR95 by about £1,450 against £3,100. At 420 / 120 the tail flights are the ones with many empty business seats, and a common shock lines them up with empty economy seats on the same days. The higher limits fill those seats, so there is less tail to line up.
- **Divergent days barely change the recommendation's tail** (P95 +£200, CVaR95 −£200) and make it slightly *more* likely to bump someone (+3 points), because a low-economy, high-business day still bumps in economy.
- **In every setting the recommendation's tail is far below today's:** P95 £24.0–24.4k against £40.1–43.7k, and CVaR95 £34–36k against £49–54k (data inputs).
- **Bumping frequency is the cost:** P(any bump) is 90–97% and P(business bump) 78%, against 60–73% and 34–37% today, in every correlation setting. Correlation does not change that trade-off.

7c. Upgrades: bumped economy passengers in spare business seats

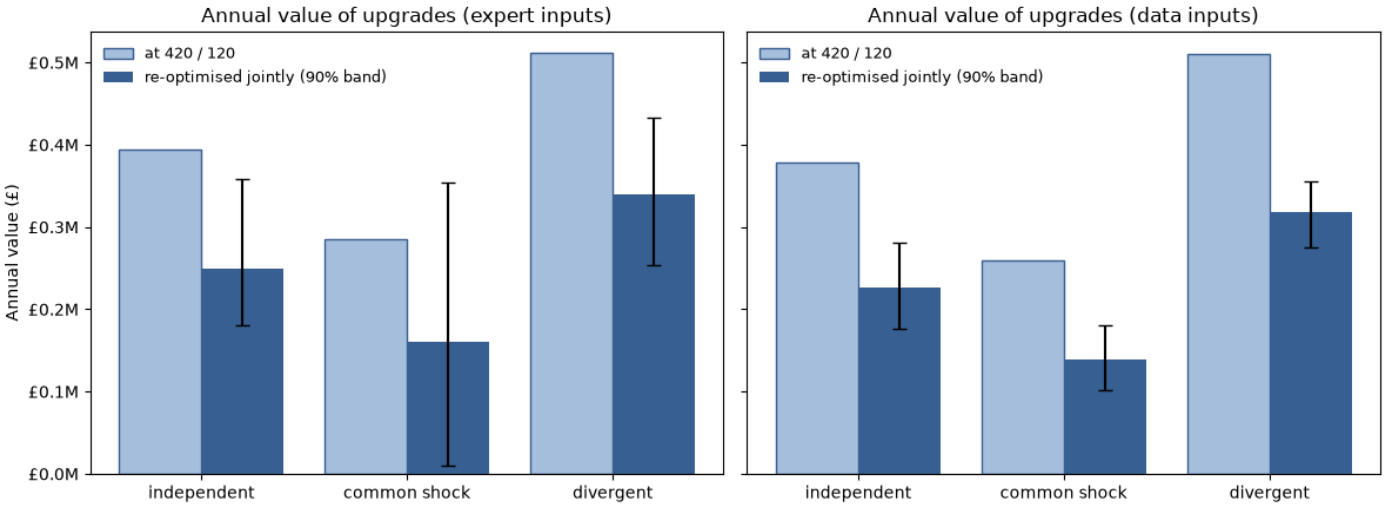
Rule modelled: on each flight, upgrades = min(economy passengers bumped, empty business seats). Each upgrade avoids the £150 economy bump cost. The business seat still earns no business fare, so its £1,450 empty-seat cost stays in. Business passengers are not moved down into economy.

The grid now has to be joint. The upgrade saving depends on both cabins on the same flight, so the Step 6 shortcut (economy cost + business cost) no longer holds. Each cabin is still swept once. Every flight's economy bumps and business empty seats are kept at every limit, and the expected number of upgrades is computed for **all 3,721 pairs** of limits, per outer draw, on the same flights. This is exact, not an approximation. It is repeated for both input sources and all three correlation settings.

Annual value of upgrades, three ways:

- **at 420 / 120:** upgrades added to today's policy;
- **same limits:** upgrades added to the best no-upgrade limits;
- **re-optimised:** best cost without upgrades minus best cost with upgrades after jointly re-optimising, with a 90% band across outer draws.

inputs	correlation	best without upgrades	cost without upgrades	best with upgrades	cost with upgrades	upgrades per flight	economy bumps left	annual value at 420 / 120	annual value, same limits	annual value, re-optimised	re-optimised 90% low	re-optimised 90% high
expert	independent	430 / 142	£9,983	431 / 141	£9,813	1.21	9.22	£394,430	£237,280	£248,697	£180,896	£358,171
	common shock	430 / 142	£9,983	431 / 142	£9,873	0.74	9.68	£284,688	£152,554	£160,995	£9,937	£353,997
	divergent	430 / 142	£9,982	431 / 141	£9,749	1.63	8.80	£511,568	£325,187	£339,914	£253,012	£432,647
data	independent	429 / 139	£9,223	429 / 139	£9,068	1.03	8.43	£379,014	£226,610	£226,610	£175,633	£280,763
	common shock	429 / 139	£9,221	429 / 139	£9,126	0.64	8.82	£259,857	£139,191	£139,191	£101,501	£180,357
	divergent	429 / 139	£9,221	429 / 138	£9,003	1.54	7.92	£509,921	£316,033	£317,673	£275,858	£355,528



What 7c shows

- **Upgrades don't change the limits.** The joint optimum is 429 / 139 (independent and common shock) or 429 / 138 (divergent) on data inputs, and 431 / 141–142 on expert inputs. Re-optimising adds almost nothing over simply adding upgrades to the Step 6 limits.
- **They are worth about £0.23M a year on data inputs** (90% band £0.18–0.28M) with independent cabins: about 1 upgrade per flight, out of about 9.5 economy bumps. That is

under 2% of the remaining cost.
- **Correlation now matters, as Step 5 anticipated.** A common shock cuts the value to **£0.14M**, because economy is overfull on the same days that business is full, so there are fewer spare seats when they are needed. Divergent days raise it to **£0.32M**, since economy is full exactly when business has seats to spare. Expert inputs give similar figures (£0.16–0.34M) with wider bands.
- **Upgrades are worth more under today's policy** (£0.26–0.51M), because 420 / 120 leaves more business seats empty. As the recommendation fills those seats, the upgrade option shrinks.
- **The verdict:** formalise the practice, since it is free money and staff already do it. But it is a bonus on top of the new limits, not a reason to choose different ones. Two cautions: the log can't tell us how often it already happens (so part of this value may already be in the actual costs), and a known upgrade route could change economy passengers' behaviour.

7d. Regret table

Policies (all limits fixed in advance):

1. current 420 / 120;
2. best on expert inputs (Step 6);
3. best on data inputs (Step 6);
4. best with upgrades: the 7c joint optimum on data / independent, **with upgrades used in every world**;
5. weekday / weekend split: the data economy limit with Step 6's business limits of 134 on weekdays and 150 at weekends.

Worlds: each one fixes the true mean no-show rates, so the epistemic uncertainty is switched off ("what if we knew the truth?"). The per-flight spread and the common random numbers are the same as before.

- **Expert low / mode / high:** the experts' 3% / 5% / 8% economy and 15% / 20% / 30% business, every day.
- **Data mean:** the ordinary-day means from Step 4, with business split by weekday (18.2%) and weekend (26.1%) and each day type's own spread. This lets the split policy show its value.
- **Bump cost × 3:** the data-mean world at £450 / £900.
- **Common shock:** the data-mean world with correlated cabins (+0.4).
- **Thanksgiving-type divergent:** the mean rates actually logged in Thanksgiving week (24–30 Nov), with divergent cabins (−0.4) and the ordinary-day spread.

Regret = a policy's expected cost per flight in a world minus the cheapest of the five policies in that world. Worst-case regret is the largest across the seven worlds. A second table shows the best single limits (no upgrades, no split) with hindsight in each world, on a wider grid (economy 400–480, business 100–220), as a check on how far the five candidates are from what was possible.

	economy no-show	business no-show, weekdays	business no-show, weekends	correlation	bump cost ×
expert low	3.0%	15.0%	15.0%	independent	1
expert mode	5.0%	20.0%	20.0%	independent	1
expert high	8.0%	30.0%	30.0%	independent	1
data mean	5.2%	18.2%	26.1%	independent	1
bump cost × 3	5.2%	18.2%	26.1%	independent	3
common shock	5.2%	18.2%	26.1%	common shock	1
Thanksgiving-type divergent	1.5%	36.5%	36.5%	divergent	1

	expert low	expert mode	expert high	data mean	bump cost × 3	common shock	Thanksgiving-type divergent	worst-case regret	worst-case regret, annual
best with upgrades 429 / 139	£832	£0	£1,013	£347	£2,101	£401	£0	£2,101	£3,068,040
split 429 / 134 weekdays, 150 weekends	£1,035	£512	£1,963	£0	£1,592	£0	£2,456	£2,456	£3,586,284
best on data 429 / 139	£908	£146	£1,219	£500	£2,561	£500	£1,722	£2,561	£3,739,630
best on expert 430 / 142	£1,532	£281	£0	£564	£4,049	£564	£169	£4,049	£5,911,470
current 420 / 120	£0	£4,581	£15,205	£5,557	£0	£5,561	£14,742	£15,205	£22,199,944

world	hindsight economy	hindsight business	hindsight cost	best candidate	best candidate cost	gap to hindsight
expert low	417	128	£7,467	current 420 / 120	£8,378	£911
expert mode	428	138	£9,038	best with upgrades 429 / 139	£8,909	−£129
expert high	444	162	£11,667	best on expert 430 / 142	£15,563	£3,896
data mean	429	139	£9,145	split 429 / 134 weekdays, 150 weekends	£8,645	−£500
bump cost × 3	419	127	£15,738	current 420 / 120	£16,911	£1,173
common shock	429	139	£9,144	split 429 / 134 weekdays, 150 weekends	£8,645	−£500
Thanksgiving-type divergent	409	182	£11,130	best with upgrades 429 / 139	£22,254	£11,124

What 7d shows

- **Today's policy is only right if bumping is much dearer or no-shows sit at the experts' low end.** It has zero regret in the *expert low* and *bump cost × 3* worlds, and £4.6k–£15.2k per flight in every other world. Its worst case is **£15.2k per flight (£22.2M a year)** if no-shows are at the experts' high end.
- **All four new policies cut the worst-case regret by a factor of 4–7.** The minimax-regret choice is **429 / 139 with upgrades**: its worst case is £2.1k per flight (£3.1M a year), in the *bump cost × 3* world. Without upgrades the same limits have a worst case of £2.6k (£3.7M a year). The expert-based 430 / 142 is clearly worse (£4.0k), because it overbooks more and so suffers more when bumping gets dearer.
- **Bump cost × 3 is the world that hurts every new policy most.** This matches 7a: ×3 is past the ×2.3 break-even. The bump cost is the one input that could make the change a mistake.
- **The split policy wins whenever weekdays and weekends really differ** (data-mean and common-shock worlds): about £500 per flight better than 429 / 139 alone, and £350 better than 429 / 139 with upgrades. But its weekend limit of 150 is costly when there is no weekend effect, or in a Thanksgiving-type week (worst case £2.5k). Combining the split with upgrades is the natural next candidate.
- **A Thanksgiving-type week defeats every fixed policy.** With hindsight the right limits that week are about **409 / 182**, at £11.1k per flight, against £22.3k for the best candidate. Holiday and disruption weeks need their own override, set from bookings data for those dates, rather than a better year-round number.
- **Negative gaps in the hindsight table** are not errors. The hindsight limits are a single pair with no upgrades and no split, so policies 4 and 5 can beat them.

7e. Sensitivity: which input matters most?

One input at a time is moved to the ends of a plausible range, with everything else at the base case: the ordinary-day data means (economy 5.2%, business 20.6% every day), independent cabins, and £150 / £300 bump costs. For each end the table shows:

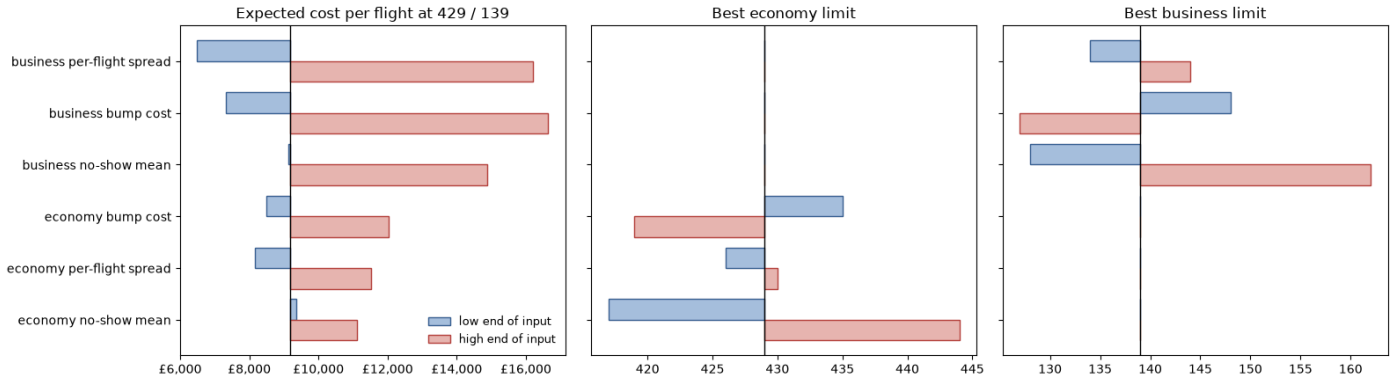
- the expected cost per flight of the **fixed** recommendation 429 / 139;
- the **best limits** for that input value;
- the **regret** of staying at 429 / 139 instead of moving to those limits.

Input	Low	High	Why this range
Economy / business no-show mean	3% / 15%	8% / 30%	The experts' full range
Economy / business bump cost	×0.5	×3	×3 is the regret-table world, beyond the 7a break-even
Economy / business per-flight spread	SD ≈ ×0.5	SD ≈ ×2	Beta concentration divided by (SD multiplier)², i.e. half or double the flight-to-flight variation fitted in Step 4

Correlation is left out on purpose. Without upgrades it has no effect on the expected cost (7b), and with upgrades it moves the cost by only about ±£0.1M a year (7c).

Base case (data means, base spread and bump costs): cost at 429 / 139 £9,190 per flight; best limits 429 / 139

input	low value	high value	cost at low	cost at high	cost swing	best limits at low	best limits at high	regret at low	regret at high
business per-flight spread	SD ≈ ×0.5	SD ≈ ×2.0	£6,493	£16,216	£9,723	429 / 134	429 / 144	£282	£107
business bump cost	£150	£900	£7,327	£16,638	£9,310	429 / 148	429 / 127	£407	£2,611
business no-show mean	15%	30%	£9,123	£14,869	£5,746	429 / 128	429 / 162	£1,035	£3,854
economy bump cost	£75	£450	£8,482	£12,019	£3,537	435 / 139	419 / 139	£144	£1,155
economy per-flight spread	SD ≈ ×0.5	SD ≈ ×2.0	£8,150	£11,522	£3,371	426 / 139	430 / 139	£59	£1
economy no-show mean	3%	8%	£9,352	£11,102	£1,750	417 / 139	444 / 139	£784	£1,262



What 7e shows

- **Business inputs dominate the cost.** The business per-flight spread (£6.5k–£16.2k per flight), the business bump cost (£7.3k–£16.6k) and the business no-show mean (£9.1k–£14.9k) are the three widest bars. All three economy inputs swing the cost by less than £3.6k.
- **What moves the cost is not always what moves the decision.** The per-flight spread is the top cost driver but shifts the best limits only a little (business 134–144, economy 426–430), so getting it wrong changes the budget, not the policy: regret is under £300 per flight.
- **The decision is driven by the means and the bump costs.** The best business limit runs from 128 to 162 across the business no-show range and from 148 to 127 across the business bump-cost range. The best economy limit runs from 417 to 444 across the

economy no-show range and from 435 to 419 across the economy bump-cost range. Each input moves only its own cabin's limit, because without upgrades the cabins are costed separately.
- **The biggest regrets come from a business no-show mean at 30%** (£3.9k per flight: the limit should be 162) and **a business bump cost at ×3** (£2.6k: the limit should be 127). These are the two numbers most worth pinning down. The first is already well measured: the day-level CI from Step 4 is about ±1.5–2 points, not 15–30%. The second, the business goodwill cost, has never been measured at all.

What Step 7 shows

- **The recommendation is robust to everything except the bump cost.** It holds across expert vs. data inputs, all three correlation settings and the upgrade question. It stops beating 420 / 120 if bump costs rise to about **2.3× today's estimate (about £350 / £700)**, and the competitor's policy is exactly what could cause that.
- **On minimax regret, 429 / 139 with upgrades formalised is the best of the five candidates.** Its worst case is £2.1k per flight, against £15.2k for today's policy. Upgrades themselves are a small bonus (about £0.14–0.32M a year, depending on correlation) and don't change the limits.
- **Correlation matters for risk, not the average.** The recommendation's tail is less exposed to common shocks than today's policy, and stays far below it in every setting.
- **Two things a single number can't handle:** (1) a bump-cost rule, i.e. the limits table from 7a as a lookup to update once the goodwill cost is re-estimated, and (2) a holiday/disruption override, since no fixed policy comes close in a Thanksgiving-type week.
- **The data most worth collecting:** the true goodwill cost of a bump, especially in business, ahead of any further refinement of the no-show model.

Step 8: The no-refund question

Finance wants to stop refunding no-shows. This step switches from cost to **profit per flight**:

- **With refunds (today):** fare × passengers flown − bump cost × passengers bumped. This is the full-plane revenue minus the Step 1 cost, so the two objectives agree.
- **Without refunds:** the same, **plus the fares that no-shows keep paying**: fare × bookings × no-show rate.

Dropping refunds can change behaviour in two ways, and neither can be measured from the log. So both are set on a grid:

Input	Values	How it enters
No-show rate	100% / 75% / 50% / 25% of today's	Scales the mean no-show rate in both cabins (ordinary-day means from Step 4: 5.2% / 20.6%). The per-flight spread (beta concentration) is kept.
Demand loss	0 / 5 / 10 / 20%	Fewer people book: bookings = min(limit, demand), with demand cut by the same share in both cabins.

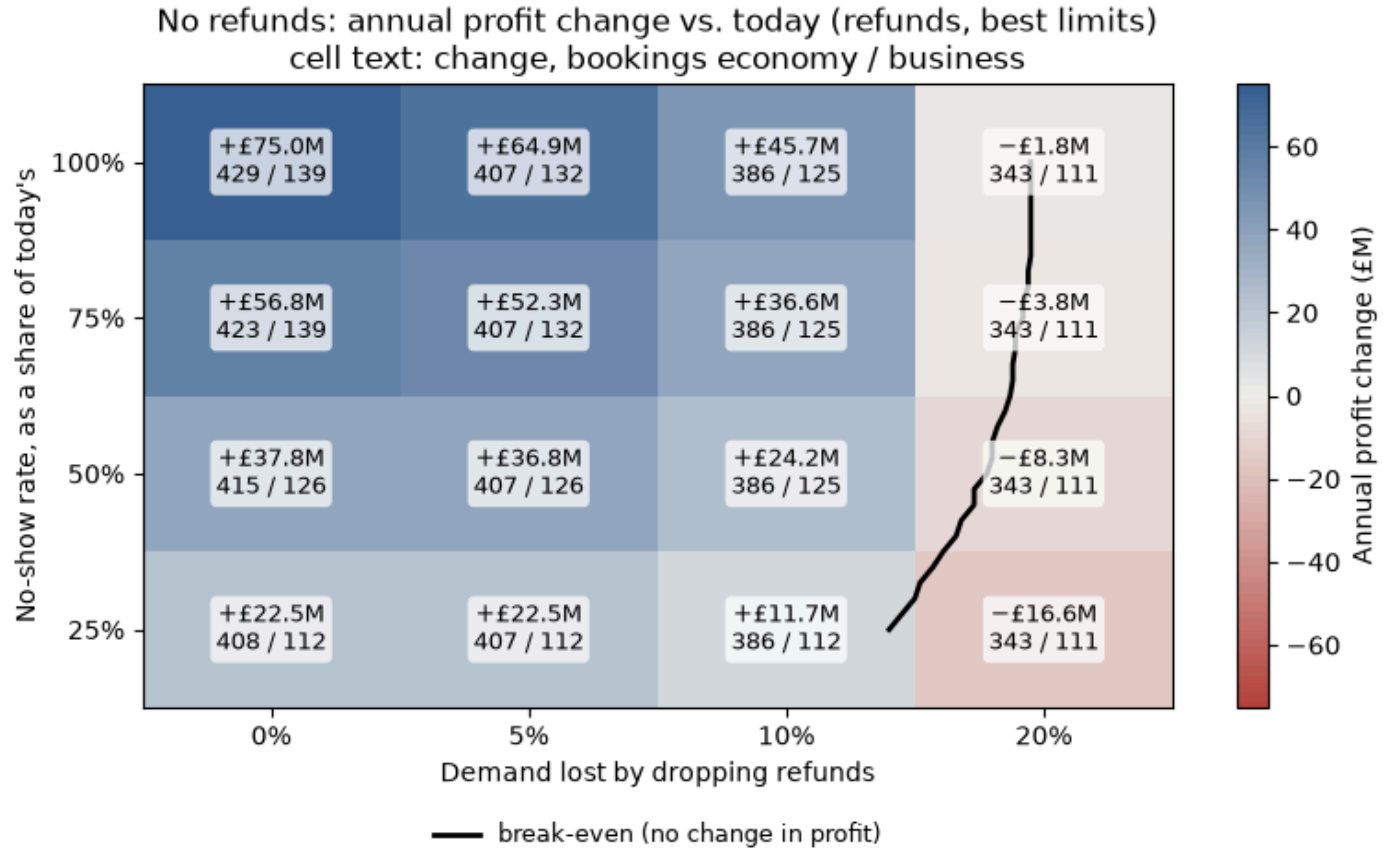
In each cell the booking limits are **re-optimised** for that cell.

Assumptions to keep in view:

- **"Today" is refunds with the limits re-optimised in the same model (429 / 139), not 420 / 120 literally.** Comparing with 420 / 120 would mix the refund decision with the re-optimisation gain from Step 6. The first cell prints the amount to add to every cell for a comparison with 420 / 120.
- **Today's demand is set exactly at the refund-optimal limits (`DEMAND_HEADROOM = 0`).** The log can't show true demand, because bookings hit the limit on every flight. So any demand loss costs bookings straight away. This is the cautious case. If demand really runs well above the limits, a small loss costs nothing.
- **Exact expectations.** With the mean no-show rate fixed in each cell (as in 7d and 7e), show-ups are beta-binomial, so the expected profit is computed exactly rather than simulated. The first cell checks the result against the 7e simulation.

```
Today (refunds, best limits 429 / 139): profit £315,814 per flight, i.e. cost £9,186
Check against the 7e simulation at 429 / 139: cost £9,190, best limits 429 / 139
Literal 420 / 120 with refunds: £7.3M a year below that, so add this to every cell below to compare against 420 / 120 itself
```

no-show rate	demand loss	economy bookings	business bookings	profit	change per flight	annual change
100%	0%	429	139	£367,211	£51,397	£75,039,766
	5%	407	132	£360,271	£44,457	£64,907,524
	10%	386	125	£347,112	£31,298	£45,694,530
	20%	343	111	£314,549	£-1,265	£-1,846,503
75%	0%	423	139	£354,722	£38,908	£56,805,042
	5%	407	132	£351,619	£35,804	£52,274,500
	10%	386	125	£340,895	£25,081	£36,617,876
	20%	343	111	£313,179	£-2,636	£-3,848,069
50%	0%	415	126	£341,727	£25,913	£37,833,125
	5%	407	126	£340,995	£25,180	£36,763,519
	10%	386	125	£332,409	£16,595	£24,228,820
	20%	343	111	£310,117	£-5,697	£-8,317,880
25%	0%	408	112	£331,226	£15,412	£22,500,854
	5%	407	112	£331,223	£15,409	£22,497,619
	10%	386	112	£323,823	£8,008	£11,692,392
	20%	343	111	£304,464	£-11,350	£-16,571,095



What Step 8 shows

- **Demand loss decides the answer; the no-show response barely moves it.** Dropping refunds raises profit as long as it loses less than about **14–19.5% of bookings**. The break-even is 19.5% if no-show rates stay the same and falls to about 14% if they drop to a quarter of today's. Every cell up to 10% demand loss is a gain, and every cell at 20% is a loss.
- **The headline gain is large, but only if no one reacts.** With unchanged behaviour, the kept no-show fares are worth about **£75M a year** (£51k per flight), mostly from business: 20.6% of 139 bookings at £1,450. That cell is the least likely: a no-refund ticket changes the product.
- **Reading the cells:** the second line is bookings per flight, economy / business. Where it is below the refund-optimal 429 / 139, demand is the binding constraint and any limit at or above it books the same; a figure below 400 (e.g. 343) means economy flies with empty seats however the limit is set.
- **Fewer no-shows shrink the gain but don't wipe it out.** Less fare is kept, but fewer seats fly empty and the limits come down (to about 408 / 112 at a quarter of today's no-shows), which cuts bumping. At 5% demand loss the gain is still £22–65M a year across the no-show range.
- **Business no-shows are worth more than a business bump costs.** Without refunds and with today's no-show rate, another business booking is worth 20.6% × £1,450 ≈ £299 from a no-show, against 79.4% × £300 ≈ £238 from a bump. So the model would overbook business without limit, and only the demand cap stops it. This means the £300 business bump cost is too low for a heavily overbooked, non-refundable business cabin. A passenger who paid £1,450 non-refundable and then gets bumped will be angrier than today's.
- **The literal 420 / 120 comparison** adds about £7.3M a year to every cell. That is the Step 6 re-optimisation gain, not a refund effect.

What to check before trusting either answer:

1. **Demand loss by cabin.** This is the number that decides it, and the log can't show it, because bookings are always at the limit. Business travellers are the ones paying for flexibility, so business may lose far more than 20%.
2. **Real demand headroom.** Rerun with `DEMAND_HEADROOM` above 0 if the booking system shows requests turned away at the limit. Headroom pushes the break-even line to the right.
3. **What the competition does.** If competitors keep refunds, part of the demand loss is our customers switching to them, and it lands on the route's most valuable segment.
4. **A test rather than a switch.** The alternative to an all-or-nothing change is to sell a cheaper non-refundable fare **alongside** the refundable one. That measures demand loss and the no-show response directly, and it is how many full-service carriers already handle it.

Step 9: Summary for the report

One table comparing the three policies the report discusses, on the base case: **data inputs, independent cabins**, the same simulated flights as Steps 5–7.

- **Current:** 420 / 120.
- **Recommended:** the Step 6 limits (`SUMMARY_LIMITS` , set to `RECOMMENDED`). Change that one line if the report recommends other limits; everything below follows.
- **Recommended + upgrades:** the same limits, with bumped economy passengers moved into empty business seats when there are any (the 7c rule: the business seat still earns no business fare, and the economy bump cost is avoided).

Annual figures are per flight × 4 flights × 365 days. Savings are against 420 / 120, computed per outer draw on the same flights, so the 90% band and P(saving > 0) reflect epistemic uncertainty only. The break-even bump cost is the multiplier on £150 / £300 at which the policy stops beating 420 / 120 (as in 7a). Worst-case regret comes from the 7d table, so it is measured against the cheapest of the five 7d candidates in each world.

The table is saved to `report/policy_summary.csv` as raw numbers, metrics as rows and policies as columns.

metric	current	recommended	recommended + upgrades
economy limit	420	429	429
business limit	120	139	139
upgrades	False	False	True
expected cost per flight (£)	£14,261	£9,223	£9,068
expected cost 90% band low (£)	£12,682	£8,843	£8,707
expected cost 90% band high (£)	£15,816	£9,631	£9,470
SD of cost per flight (£)	£13,276	£7,467	£7,198
P95 cost per flight (£)	£41,550	£23,950	£23,200
CVaR95 cost per flight (£)	£51,147	£34,353	£33,098
P(any bump)	68.2%	94.2%	88.6%
P(business bump)	36.9%	78.1%	78.1%
economy bumps per flight	3.91	9.46	8.43
business bumps per flight	2.50	12.42	12.42
upgrades per flight	0.00	0.00	1.03
load factor	97.4%	99.1%	99.3%
annual cost (£)	£20,820,755	£13,466,065	£13,239,455
annual saving vs 420 / 120 (£)	£0	£7,354,690	£7,581,300
annual saving 90% band low (£)	£0	£5,470,169	£5,647,209
annual saving 90% band high (£)	£0	£9,191,669	£9,448,956
P(saving > 0)		100.0%	100.0%
passengers bumped per year	9,364	31,947	30,436
upgrades per year	0	0	1,511
break-even bump cost ×		2.32	2.42
worst-case regret per flight, 7d (£)	£15,205	£2,561	£2,101

```
Saved for the report:
  report/policy_summary.csv   Step 9: policy summary table
  report/figures/step3b-noshow-distribution.png   Step 3b: no-show rate per flight vs. the expert range
  report/figures/step3b-weekly-noshow.png   Step 3b: weekly mean no-show rate
  report/figures/step5-cost-distribution-current.png   Step 5: cost per flight at 420 / 120, expert vs. data inputs
  report/figures/step6-cost-curves.png   Step 6: expected cost vs. booking limit, with epistemic bands
  report/figures/step6-business-weekday-weekend.png   Step 6: business cost by day type
  report/figures/step7a-bump-cost.png   Step 7a: best limits and annual saving vs. bump cost
  report/figures/step7c-upgrade-value.png   Step 7c: annual value of upgrades
  report/figures/step7e-tornado.png   Step 7e: sensitivity tornado
  report/figures/step8-no-refund-heatmap.png   Step 8: no-refund profit change, with break-even line
```